

Past

Paper

Adv

Design

6th

Semster:-

~(2017)~

Question No: 02.

~(9 marks)~

~(i)~

Fixed Effect:-

(i) K treatment are selected by choice of experiment.

(ii) α_j is fixed

(iii) α_j is estimated

(iv) $H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_K$

(v) Conclusion are only for selected treatment

Random effect:-

(i) ' K ' treatment are selected by random process from the population of treatment.

(ii) α_j is a random variable

(iii) Variability of treatment is estimated

Say σ_α^2

(iv) $H_0: \sigma_\alpha^2 = 0$

(v) Conclusion are application for all the treatment of population.

Mixed effect:- Same treatment are selected by choice and treatment are selected randomly is called mixed effect model. It cannot use in CR but use in RCB and LSD Design:-

Q (ii)

(6 mark)

Complete Confounding:-

When a Confounding variable is perfectly correlated with the treatment variable.

- Means that the Confounding variable is identical for all units within a treatment group.
- Results in a complete loss of information about the treatment effect.

Example:-

In a study on the effect of a new drug if all the patients receiving the drug are male and all patients receiving the placebo are female. gender is completely confounded with the treatment.

Partial Confounding:-

When a Confounding variable is correlated with the treatment variable but not perfectly.

- Means that the confounding variable has some variation within treatment group.
- Results in a partial loss of information about the treatment effect.

Example:-

In a study on the effect of exercise on weight loss. If participants who exercise regularly tend to be younger than those who don't.

Balance

- Each number
- Each in the of exam

4

two

•

the

•

in

of

ex

m

a

Balanced incomplete Block Design:-

- Each treatment is replicated the same number of times in each block
- Each pair of treatment appears together in the same block an equal number of times

example:-

4 treatment (A, B, C, D) in 4 block, with each treatment appearing twice in each block.

partially Balanced Incomplete Block design:-

- each treatment is not replicated the same number of times in each block.
- each pair of treatment appears together in the same block an different number of times.

example:-

4 treatment (A, B, C, D) in 4 block, with A and B appearing together twice, but C and D only once.

Question No. 03

Given the following Anova table

Source	df	Mean Square
A	2	5341.86
B	2	19558.36
AB	4	2403.44
Error	27	675.21
Total	35	

Assuming completely randomization model interpret the effect of A, B, and AB when

- The level of both factor are fixed
- The level of both factor are random
- levels of A are fixed but level of B are random

Solution:-

(i)

Assuming both factor (A and B) are fixed

Factor A (A):-

$$MSA = \frac{SSA}{dFA} = \frac{5341.86}{2}$$

$$= 2670.93$$

$$F_A = \frac{MS_A}{MS_{Error}} = \frac{2670.93}{675.21}$$

$$= 3.9557$$

This F_A value tests whether there is a significant difference among the means of factor A levels.

Factor B (B):

$$MS_B = \frac{SS_B}{df_B} = \frac{19558.36}{2}$$

$$= 9779.18$$

$$F_B = \frac{MS_B}{MS_{Error}}$$

$$= \frac{9779.18}{675.18}$$

This F_B value tests whether there is no significant difference among the means of factor B levels.

Interaction (AB):-

$$MS_{AB} = \frac{MS_{AB}}{df_{AB}} = \frac{2403.44}{4} \\ = 600.86$$

This F_{AB} value tests whether the interaction between A and B is significant.

~~Q2)~~

Assuming level of A and B are random.

The Anova table provided does not directly support this assumption.

for random factor,

one would typically use a mixed-effect model not Anova.

~~Q3)~~

Assuming level of A is fixed but level of B random:-

Factor A (A):-

Same as in

Scenario

factor B (B):-

In a mixed-effect model context, the interpretation changes. MS_B would represent the variance component associated with random factor B.

Interaction (AB):-

MS_{AB} would also be interpreted in the context of covariance components under mixed effect model.

Question No: 04

Notes pg 186 to 190

* Question No: 05

Complete the following table

Source	df	SS	MS	F
Block	3	28.44	-	-
A	-	4.92	1.64	-
Error (a)	-	-	-	-
Subtotal	-	40.85	-	-
B	-	29.30	-	-
AB	9	-	-	-
Error (b)	-	86.65	-	-

Complete Anova table and test the significance among the treatment means.

Solution:-

SoV	df	SS	MS	F
Block	$(a-1)(b-1) = 3$	28.44	9.48	11.42
A	$(a-1)(b-1) = 3$	4.92	1.64	1.975
Error(a)	9	7.49	0.83	
Subtotal	63	40.85	0.65	
B	4	29.30	9.76	4.05
AB	9	6.63	0.73	0.30
Error(b)	$\frac{4(a-1)(b-1)}{4(3)(3)} = 36$	86.65	2.406	

$$r = 4, a = 4, b = 4$$

$$\begin{aligned} \text{Error(a)} &= \text{Subtotal} - \text{Block} - A \\ &= 40.85 - 28.44 - 4.92 = 7.49 \end{aligned}$$

$$\begin{aligned} SS_{AB} &= \text{Subtotal} - A - B \\ &= 40.85 - 4.92 - 29.30 = 6.63 \end{aligned}$$

Step 1:-

H_0 : All treatment are ~~treatment~~ equal

H_1 : all treatment are not equal

Step 2:-

$$\alpha = 0.05$$

Step 3:-

$$F = \frac{MS_{\text{effal}}}{MS_{\text{error}}}$$

$$F_{0.05(3,9)} = \cancel{3.86} 3.86$$

$$3.86 > 11.394$$

We conclude that treatment mean are equal We do not reject H_0

Question

The mil additive car as constr block with the a draw

Addi

1

2

3

4

5

Solu

Question : 05

An engineer is studying the mileage performance of 5 type gasoline additive. In the road test he wishes to use car as blocks however, because of a time constraint, he must use an incomplete block design. He runs the balanced design with the 5 block that follows. Analyze the data from this experiment and draw conclusions.

Additive	Car				
	1	2	3	4	5
1	-	17	14	13	12
2	14	14	-	13	10
3	12	-	13	12	11
4	13	11	11	12	-
5	11	12	10	-	10

Solution:-

Notes

df $\rightarrow$ 212 to 215

~(2018)~

Question NO: 02

~(ii)~ Factorial Design:-

Factorial design is a statistical technique used to experimentally design and analyze data from experiments that involves multiple variables or factors. It's commonly used in field like science engineering and finance to identify the relationships between variables and their effects on outcomes.

~(ii)~ Confounding:-

Confounding is a common issue that can lead to biased estimates and incorrect interpretations. To address confounding, researchers use various techniques, such as matching, stratification, and regression analysis, to control for the effect of confounding variables.

Example:-

Imagine you want to know if ice cream makes people happy

Youden Square Design

A Youden Square is an experimental design used in statistics to analyze the effect of multiple variables (factors) on a response variable. It's a type of fractional factorial design that helps to identify the most important factors and their interaction.

It's commonly used in fields like quality control engineering and scientific research.

(iv)

Incompelet Block Design

An

incompelet block design is a statistical technique used to analyse data from experiment where

- (1) There are multiple experiments to be compared
- (2) The treatment are applied to small group or blocks
- (3) Not all treatments applied to every block.

Question No: 03

Suppose a study is performed to determine the effect of 3 factors A, B, C. Suppose that each batch of raw material is only large enough to test only 4 treatment combinations. Thus each replicate of 2^3 design must be run in 2 blocks. Two replicates are run with ABC confounded in replicate-I and AB confounded in replicate-II. The data is as follows.

Replicate-I		Replicate-II	
ABC Confounded		AB Confounded	
Block 1	Block 2	Block I	Block 2
(1) = 3	a = 0	(1) = 1	a = 1
ab = 2	b = 1	C = 0	b = 0
ac = 2	C = 1	ab = 3	ac = 1
bc = 1	abc = 6	abc = 5	bc = 1

Solution:-

Pg No 199 to 202..

Question No: 04

Consider the 2^5 design in four blocks of eight run each with ACDE and BCE as independent effects chosen to be confounded with block. Generate the design. Find the other effects confounded with block.

Solution

E.
(2) C
be C
and
(3) D
(4) gen
b

Block

Col
block

(1)
block

abd
b

C, ac
b

the

Solution:-

- (1) Write down the 5 factors: A, B, C, D, E.
- (2) Choose two independent effects to be confounded with the block ACDE and BCE
- (3) Determine the generator: Block = ACDE = BCE
- (4) generate the 16 runs for each block.

Block 1:

(1) a, b, ab, c, ac, bc, abc, d, ad, bd, abd, cd, acd, bcd, abcd

Block 2:

(1) a, b, ab, c, ac, bc, abc, d, ad, bd, abd, cd, acd, bcd, abcd

Block 3:

(1) a, b, ab, c, ac, bc, abc, d, ad, bd, abd, cd, acd, bcd, abcd

Block 4:

(1) a, b, ab, c, ac, bc, abc, d, ad, bd, abd, cd, acd, bcd, abcd

The other effects confounded with the block are:-

ABDE (confounded with Block 1 and 3)

ACDE (" " 2 and 4)

BCE (con " 1 and 2)

BDE (" " 3 and 4)

Block 1: $ACDE = BCE = ABDE$
 Block 2: $ACDE = BCE = -ABDE$
 Block 3: $ACDE = -BCE = -ABDE$
 Block 4: $ACDE = -BCE = ABDE$

Another solution on pg 176 to 179

Question NO: 05

Complete and Analyze the following Anova table.

S.O.V	d.f	SS	MS	F-Ratio
Varieties(V)	2	0.1781	0.08905	0.5637
blocks	5	4.1499	0.82	0.6929
error (a)	10	1.3622	0.136	-
Dates(D)	$4-1=3$	1.9622	0.654	2.3397
VXD	$(3 \times 2)=6$	0.2105	0.0351	0.12554
Error(b)	45	1.2583	0.0299	-

$$a = 3$$

$$b = 4$$

$$r = 6$$

Step 1:-

- H_0 : The effect of variate is zero.
- H_0 : The effect of date is zero.
- H_0 : The effect of block is zero.
- H_0 : The effect of interaction is zero.
- H_1 : At least one of them are not zero.

Step 2:-

$$\alpha = 0.05$$

Step 3:-

$$F = 2.3397$$

Step 4:-

Fo

Step 5:-

We

and d

So

In

full

Qu

parti
Design

B.

Step 2:-

$$\alpha = 0.05$$

Step 3:-

$$F = \frac{ms_{\text{effect}}}{ms_{\text{error}}}$$

Step 4:-

$$F_{0.05(df)} = 4.10, 3.33, 2.30, 2.80$$

We conclude that block and treatment fall in critical region. So we reject H_0 and (V) does not fall. We do not reject H_0 .

Question No: 06

Consider the following partially balanced incomplete block design.

Block	Treatment combination		
1	1	2	3
2	3	4	5
3	2	5	6
4	1	2	4
5	3	4	6
6	1	5	6

Verify the following relationship among the parameters.

$$p_{11} + p_{12}^2 = n_1$$

$$n_1 p_{12} = n_2 p_{11}^2$$

$$p_{21} + p_{22}^2 = n_2$$

$$n_2 p_{11} = n_1 p_{12}^2$$

$$p_{11} + p_{12} = n_1 - 1$$

$$n_1 p_{11} + n_2 p_{12} = r(k-1)$$

$$p_{21} + p_{22}^2 = n_2 - 1$$

$$n_1 + n_2 = a - 1$$

Solution:-

Nate & pg 228 to 234) on

(2019)

Question NO=02.

(a)

**Advantage and DisAdvantage
of Fectorial Design:-**

Advantage:-

Efficient use of Resources:-

Fectorial design allow researchers to examine the effect of multiple factors in a single experiment, thus maximizing the use of resources such as time, money, and participant.

Statistical Efficiency:-

Factorial designs often have greater statistical power compared to other designs because they enable the detection of main effects and interactions simultaneously.

Disadvantage:-

Increase Complexity:-

Using multiple independent variables can interpret results more complexly, mainly when interaction effects are present.

Limited generalizability:-

The specific condition of it may not be generalizable to the other contents.

~***~

Factorial replication:-

Factorial replication is a statistical technique used in experimental design to reduce the number of experimental runs required in a full factorial design. It involves selecting a subset of the possible treatment combinations to be run, rather than running all possible combinations.

Example:-

A 3 factor experiment with 2 level per factor would required $2^3 = 8$ runs. However, with fractionated replication you can select a fraction of these runs such as $\frac{1}{2}$, or $\frac{1}{4}$ to reduced the number of experiment runs.

(iii)

Quasi Latin Square:-

A Latin Square design is an experimental design that combines elements of Latin Square design and incomplete Block design. It's used to analyze the effect of multiple factors on a response variable while minimizing the number of experimental runs.

Uses:-

- There are many treatment to be compared
- Resource are limited (time, money, materials)
- The number of block is limited

Split-split plot design:-

Split-split plot design are type of experimental design used to evaluate the effects of multiple factors on a response variable. In a Split plot design, the experimental units are divided into large groups called whole plots, and each whole plot is divided into smaller groups called subplots and also called split plot design.

Example:-

Suppose we want to investigate the effect of two varieties of wheat (variety A and variety B) and three different dates of planting (Date 1, Date 2, and Date 3) on wheat yield.

Question No: 03

Complete Anova Table for the following factorial experiment. ABC is completely Confounded.

Replicate 1		Replicate 2	
Block 1	Block 2	Block 1	Block 2
b = 4	d = 10	a = 6	b = 6
a = 6	ab = 7	ac = 9	ac = 8
c = 9	bc = 8	c = 8	d = 11
abc = 5	ac = 6	b = 7	ab = 8

Complete the ANOVA Table for the following factorial Experiment. Draw conclusions.

Rep I		Rep II	
Block 1	Block 2	Block 1	Block 2
b=4	11=10	a=6	bc=6
a=6	ab=7	abc=9	ac=6
c=9	bc=8	c=8	11=11
abc=5	ac=6	b=7	ab=8
24	31	30	31

$$G = 116$$

$$C.F = \frac{(116)^2}{16} = 841$$

$$\begin{aligned} \text{Total SS} &= \sum \sum y_{ij}^2 - C.F \\ &= (4)^2 + (6)^2 + \dots + (8)^2 - 841 \\ &= 894 - 841 \\ &= 53 \end{aligned}$$

Subtable Rep x Blocks

Rep	Block	
	I	II
I	24	30
II	31	31
	55	61
	116	

$$SS_{\text{Blocks}} = \frac{(54)^2 + (62)^2}{8} - 841 = 4.00$$

$$SS_{\text{Rep}} = \frac{(55)^2 + (61)^2}{8} - 841 = 2.25$$

$$TSS_{\text{Rep x Block}} = \frac{(24)^2 + (30)^2 + (31)^2 + (31)^2}{4} - 841 = 8.50$$

$$\begin{aligned} \text{Error 1} &= TSS_{\text{Rep x Block}} - SS_{\text{Rep}} - SS_{\text{Rep}} \\ &= 8.50 - 4.00 - 2.25 = 2.25 \end{aligned}$$

T.C	Res	C ₁	C ₂	C ₃	SS = $\frac{(C_3)^2}{16}$
(1)	21	33	59	116	841
a	12	26	57	-10	6.25
b	11	29	-5	-8	4.00
ab	15	28	-5	18	20.25
c	17	-9	-7	-2	0.25
ac	12	4	-1	0	0.00
bc	14	-5	13	6	2.25
abc	14	0	5	-8	4.00

$$\begin{aligned} SS_{E2} &= TSS - SS_{\text{Rep}} - SS_{\text{Rep}} \\ &= SS_{E1} - SS_A - SS_B \\ &= SS_{AB} - SS_C - SS_{AC} \\ &= SS_{BC} \\ &= 53 - 2.25 - 4.00 - 2.25 \\ &= 6.25 \end{aligned}$$

$$= 11.50$$

ANOVA				
Source	df	SS	MS	F-value
Rep	1	2.25	2.25	1.00
Blocks	1	4.00	4.00	1.78
Error 1	1	2.25	2.25	
A	1	6.25	6.25	
B	1	4.00	4.00	
AB	1	20.25	20.25	
C	1	0.25	0.25	
AC	1	0.00	0.00	
BC	1	2.25	2.25	
Error 2	6	11.50	1.92	
Total	15	53.00		

Question NO : 04

In a split plot design the whole plot treatment (A) were then applied to split plot. The total SS, Block SS, whole plot Error SS are 822.97, 77.55 and 36.28 respectively. The sum for 3 block are given in the following table:

	B ₁	B ₂	B ₃	B ₄
A ₁	89	104	118	117
A ₂	100	117	119	126
A ₃	92	92	104	121

Complete the Anova and draw your conclusions. Also compute the standard error for the

- different between two 'A' treatment means
- different between two 'B' treatment means.

Solution:-

$$a = 3$$

$$b = 4$$

$$r = 3$$

$$N = 36$$

	B ₁	B ₂	B ₃	B	
A ₁	89	-	-	-	428
A ₂	100	-	-	-	462
A ₃	92	-	-	-	404
	281	313	341	364	1299

$$C.F = \frac{(T_{..})^2}{N}$$

$$= \frac{(1299)^2}{36} = 46872.25$$

Total SS
Block SS
Error (a)

$$SSA = \frac{1}{b}$$

$$= \frac{1}{(4)(3)}$$

$$= 120$$

$$SSB =$$

$$= \frac{1}{(3)(3)}$$

$$TSS_{AB}$$

$$= \frac{1}{3}$$

$$\text{Total SS} = 822.92$$

$$\text{Block SS} = 77.25$$

$$\text{Error (a)} = 36.28$$

$$SSA = \frac{1}{B \cdot r} \sum Y_i^2 - C \cdot F$$

$$= \frac{1}{(4)(3)} [(428)^2 + (462)^2 + (409)^2] - 46872.25$$

$$= 120.1667$$

$$SSB = \frac{1}{a \cdot r} \sum Y_{.j}^2 - C \cdot F$$

$$= \frac{1}{(3)(3)} [(281)^2 + (313)^2 + (314)^2 + (364)^2] - 46872.25$$

$$= 428.5278$$

$$TSS_{AB} = \frac{1}{r} \sum (A_i \cdot B_j) - C \cdot F$$

$$= \frac{1}{3} [189^2 + \dots + (100)^2 + \dots + (121)^2]$$

$$- 46872.25$$

$$614.25$$

$$SS_{AB} = TSS_{AB} - SSA - SSB$$

$$= 614.75 - 120.16667 - 428.5278$$

$$= 66.0553$$

$$ESS = TSS - A.SSA - SSB - \text{Block } SS \rightarrow E(a)$$

$$= 822.97 - 120.1667 - 428.5278 - 66.0553$$

$$= 77.55 - 36.28$$

$$= 94.387$$

S.O.V	df	SS	MS	F
Block	$r-1=2$	77.25	38.775	4.2750
A	$a-1=2$	120.1667	60.083	6.624
Error(a)	$(r-1)(a-1) = 4$	36.28	9.03	
B	$b-1=3$	428.52	142.84	27.23
AB	$(h-1)(b-1) = 6$	66.05	11.009	2.099
Error(b)	$a(b-1) = 18$	94.38	5.2438	
Total	$ab-1 = 35$	822.97		

$$4.2750 < 6.94 \quad \text{Insig.}$$

$$6.624 < 6.97 \quad \text{Insig.}$$

$$27.25 > 3.16 \quad \text{Sig.}$$

$$2.099 < 2.16 \quad \text{Insig.}$$

We conclude that B is false in H_0 which
 so we do not repeat H_0 which
 A and B block does not falls in
 reject region so we reject H_0 .

Question No: 05

fractional factorial Consider the 2^{8-4}
 $ACDF = ABCG = ABDH$ with generator $I = BCDE$
 find the aliases of C.

Solution:-

1. $BCDE = ACDF$
2. $ABCG = ABDH$

to find the aliases of C, we need to
 examine the generators and the defining
 relation:

$$\text{Defining relation: } I = BCDE = ACDF = ABCG = ABDH$$

using the defining relation:

$$1. C = (BCDE)/B = DE \text{ (since B is in the numerator)}$$

$$2. C = (ACDF)/A = DF \text{ (since A is in the numerator)}$$

$$3. C = (ABCG)/AB = G \text{ (since AB is in the numerator)}$$

$$4. C = (ABDH)/AB = H \text{ (since AB is in the numerator)}$$

Therefore, the aliases of C are:

- DE
- DF
- G
- H

Question NO-06

Data on screen colour difference on a television tube measured in degrees Kelvin are to be compared for four operators. On a given day only three operators can be used in the experiment. A balanced incomplete block design gave results as follows:

	Days			
Operation	Monday	Tuesday	Wednesday	Thursday
A	173	174	-	171
B	-	175	167	172
C	183	175	168	-
D	175	-	172	175

Do a complete analysis of these data and discuss your findings with regard to difference between operators

Solution:-

Φ
 A
 B
 C
 D
 Y: j

C.F

TSS

BSS

Solution:-

$$a = 4$$

$$b = 4$$

$$k = 3$$

$$r = 3$$

$$N = ar = 12$$

	M	T	W	T	$\Sigma y_{.j}$
A	173	174	-	171	518
B	-	175	167	172	514
C	183	175	168	-	526
D	125	-	172	175	522
Σy_{ij}	531	524	507	518	2080

$$C.F = \frac{(\Sigma y_{..})^2}{N} = \frac{(2080)^2}{12} = 360533.33$$

$$TSS = \Sigma \Sigma y_{ij}^2 - C.F = 360716 - 360533.33 = 182.67$$

$$BSS = \frac{1}{a} (\Sigma y_{.j}^2 - C.F) = \frac{1}{4} [(518)^2 + (514)^2 + (526)^2 + (522)^2] - 360533.33 = \frac{1}{4} [(1081680) - 360533.33] = -96113.33$$

$$CSS = \frac{1}{b} \Sigma y_{ij}^2 - C.F = \frac{1}{4} [(531)^2 + (524)^2 + (507)^2 + (518)^2] - C.F = 220477.5 - C.F$$

$$= -90055.83$$

$$\hat{n} = \frac{n(k-1)}{(n-1)} = \frac{3(3-1)}{4-1} = \frac{3(2)}{3}$$

$$= 6/3 = 2$$

$$SS_{\text{adj}} = \frac{k}{n} \sum Q_i^2$$

$$\therefore Q_i = Y_i - \bar{Y} = \frac{1}{n} \sum Y_j$$

$$Q_1 = 518 - \frac{1}{3} [531 + 524 + 518]$$

$$= 518 - \frac{1}{3} [1573] = -6.3$$

$$Q_2 = 514 - \frac{1}{3} [524 + 507 + 518]$$

$$= 514 - 516.3 = -2.3$$

$$Q_3 = 526 - \frac{1}{3} [524 + 524 + 507]$$

$$= 526 - 520.67 = 5.3$$

$$Q_4 = 522 - \frac{1}{3} [531 + 507 + 518]$$

$$= 522 - 518.67$$

$$= 3.3$$

$$Trss(ady) = \frac{k}{ah} \sum Q_i^2$$

$$= \frac{3}{4(2)} [(-6 \cdot 3)^2 + (-2 \cdot 3)^2 + (5 \cdot 3)^2 + (3 \cdot 3)^2]$$

$$= (0.375)(83.96)$$

$$= 31.485$$

$$ESS = TSS - BSS - CSS - Tr:SS$$

$$= 182.67 - (-90113.33) - (-90055.33)$$

$$- 31.485$$

$$= 99319.245$$

Source	df	SS
Treatment	$n-1=3$	31.485
RSS	$a-1=3$	-9011233
CSS	$k-1=3$	-90055.33
Error	$(a-1)(k-1)=9$	99319.845
Total		

Design Experimental

~ (2022) ~

Question No #01

(i) Factorial Experiments:-

A factorial experiment is a type experimental design used in statistics and research methodology. Each independent variable is called a factor, and each factor has multiple level, which are different values or setting of that factor. The experiments is conducted by manipulating the factors and measure the responses or outcomes.

(ii) Fractional replication:-

Fractional replication is a statistical technique

used in experimental design and data analysis. a fraction of the total project's objectives possible combination of experimental factors is used rather than the full factorial design. This approach is particularly useful when there are many experimental factors or when the experimental conditions are costly or time-consuming to replicate.

~(III)~

Confounding:-

~(repeat)~

~(IV)~

Design Resolution:-

A Design resolution is a statement that outlines the goals, constraints, and decisions made during the design process. It's a guiding framework that helps designers stay focused on the

their design project's requirements.

A typical
(2) problem is
(3) constraints

~(V)~

You

~(VI)~

PB

Balanced

it's a

experiment

of diff

on a

experiment

into

a subse

total project's objectives and ensure that their design solution align with project's requirement

A typical design resolution includes:-

- (1) problem statement
- (2) Design goals
- (3) Constraints
- (4) Design principle

~~(V)~~

Youdon Square

~ Repeat ~

~~(VI)~~

PBIBD

Ex = medicine

PBIBD stand for partially Balanced incomplete Block Design. It's a statistical design used in experiments to analyze the effects of different treatment or variable on a response variable.

In a PBIBD, the experimental units are divided into block and each block contain a subset of the treatment. The

Design is "partially balanced" because each treatment is compared with every other treatment a certain number of times, but not necessarily in every block.

Question No # 02

— (Repeat) —

Question No # 03

— (Repeat) —

Question No # 04

— (Repeat) —

~(2023)~

Question : 04

~d I ~

Merits And demerits of Factorial Experiment:-

Factorial Experiment:-

Factorial experiment are a type of E.D used to study the effects of multiple-factors on a response variable.

Merits:-

- Helps us understand how multiple things affect something
- Save time and resources
- Cost effect

Demerits:-

- can be hard to understand and analyze

- Needs a lot of resource (time, money, etc)
- Hard to interpret results.

~~(II)~~

Sign table:-

A sign table is a simple table used to summarize the Advantage (merits) and disadvantage (demerits) of a concept idea or decision. It's a tool to help visualize and organize the pros and cons.

Sign table:-

- A table with two columns, often marked with "+" (plus) and "-" (mins) sign, or "merits" and "demerits" headers.

~~(III)~~

Fractional replication:-

~~(Repeat)~~

(IV)

Partial confounding :-
— (Repeat) —

~~(V)~~ —

Split plot Design
— (Repeat) —

~~(VI)~~ —

Incomplete Block Design :-
— (Repeat) —

Question No # 02
Notes Pg # 171 to 175

Question No # 03
— (Repeat) —

Question No # 04

Repeat.